

OOP Assignment: Vehicle Rental System

Objective:

Build a Vehicle Rental Web Application using HTML, CSS, and JavaScript while implementing core OOP concepts including Encapsulation, Inheritance, Polymorphism, and Composition.

Project Requirements

1. UI Requirements (HTML + CSS)

- Create a webpage titled 'Vehicle Rental System'
- Form Fields:
 - Vehicle Name (input field)
 - Vehicle Type (Car / Bike / Truck dropdown)
 - Price Per Day (input field)
 - Add Vehicle button
- Display all vehicles as cards
- Each card must include Rent and Delete buttons

2. JavaScript OOP Requirements

A. Base Class (Vehicle)

Create a parent class 'Vehicle' with:

- constructor(name, pricePerDay)
- rent() method
- getDetails() method

B. Child Classes (Inheritance Required)

Create at least two child classes: Car and Bike

- Must use 'extends Vehicle'
- Must override the rent() method (Polymorphism)
- Add one specific property (e.g., fuelType or helmetIncluded)

C. Use of super Keyword

Inside child constructor, use: `super(name, pricePerDay)`

D. Encapsulation (Mandatory)

- Use private fields (`#pricePerDay`)
- Implement getter and setter for validation
- Price must be greater than 0

E. Polymorphism

Each vehicle type must show a different rent message:

- Car → 'Car rented with AC service'
- Bike → 'Bike rented with helmet'
- Truck → 'Truck rented for heavy load'

F. RentalSystem Manager Class (Composition)

Create a class `RentalSystem` with:

- vehicles array
- `addVehicle(vehicle)`
- `deleteVehicle(index)`
- `render()` method

Bonus Features (Optional)

- Search functionality
- Store data using `LocalStorage`
- Edit vehicle feature
- Revenue counter
- Availability status (Available / Rented)

Evaluation Criteria

HTML Structure – 10 Marks

CSS Styling – 10 Marks

OOP Structure – 20 Marks

Encapsulation – 15 Marks

Inheritance – 15 Marks

Polymorphism – 15 Marks

DOM Manipulation – 15 Marks

Reflection Questions

1. Why did we use extends?
2. Why is super() required?
3. Where did you implement polymorphism?
4. How did you implement encapsulation?
5. Could this system work without inheritance? Why or why not?